

Project ID :

25-26J-250

1. Topic (12 words max)

 Intelligent Cement Mix Design and CO₂ Emission Reduction

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

About 8% of the world's CO₂ emissions come from the cement industry, mostly from the calcination of limestone used to produce clinker [1]. Half of the 0.527 tons of CO₂ produced by this process alone per ton of clinker comes from the calcination reaction [2], [3].

Concrete crack detection by hand is frequently ineffective, time-consuming, and dependent on expert judgment [4]. In a similar vein, conventional cement mix design is not optimized, which limits its ability to lessen environmental impact [5]. Few tools combine several intelligent features into a single platform, despite the fact that Machine Learning (ML) techniques are being used to forecast strength and optimize mix designs [6].

There is potential for reducing emissions by replacing clinker with low-carbon substitutes like supplementary cementitious materials [7]. developments in carbon capture, fuel switching, and Substituting raw materials are also being studied [8]. Cement technology is increasingly utilizing ML and AI for applications such as mix optimization, strength prediction, and crack detection [9].

Convolutional Neural Networks (CNNs) have been used in computer vision for automated crack detection [10], and deep learning models such as MobileNetV2 provide improved monitoring accuracy [11]. Other machine learning algorithms, like neural networks and support vector machines, have demonstrated promise in compressive strength prediction [12].

Nevertheless, it is still uncommon to incorporate such intelligent features into unified web platforms [13]. While sophisticated optimization algorithms seek to enhance neural network performance in structural evaluations [15], recent research also looks at explainable AI (XAI) for transparent predictions [14].

There are still a number of important gaps:

- Integrated Solutions: The majority of research concentrates on discrete tasks (such as strength prediction or crack detection). Platforms that combine cement recommendations, mix optimization, and crack detection are scarce. [1], [6], [9].
- Real-time Web Applications: There aren't many portable, real-time tools that construction workers can use in the field [10], [11], [14].
- Clinker Replacement Systems: Although alternative SCMs have been found, there are few intelligent systems that can optimize replacement ratios according to construction objectives. [2], [3], [7], [8].
- Customized Cement Selection: No comprehensive system suggests cement varieties based on site-specific requirements and circumstances. [5], [6], [12].
- Practical Deployment: ML models in labs and their usability in the real world differ significantly, especially for non-technical users. [4], [9], [13], [15].

References

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- [3] B. Martinez and C. Thompson, "Carbon emissions mitigation methods for cement industry using a systems dynamics model," *Journal of Cleaner Production*, vol. 385, pp. 135-148, 2023.
- [4] K. Singh et al., "Crack detection in concrete structures using deep learning," *Sustainability*, vol. 14, no. 13, pp. 8117-8132, 2022.
- [5] L. Chen and M. Wang, "Machine learning in concrete science: applications, challenges, and best practices," *npj Computational Materials*, vol. 8, pp. 127-145, 2022.
- [6] R. Kumar et al., "Machine learning in concrete technology: A review of current researches, trends, and applications," *Frontiers in Built Environment*, vol. 9, pp. 1145591-1145612, 2023.
- [7] S. Anderson and P. Davis, "Potential of reducing CO2 emissions in cement production through altering clinker compositions," *Industrial & Engineering Chemistry Research*, vol. 63, no. 28, pp. 12456-12468, 2024.
- [8] T. Wilson et al., "Roadmap to a net-zero carbon cement sector: Strategies, innovations and policy imperatives," *Journal of Environmental Management*, vol. 365, pp. 121485-121502, 2024.
- [9] U. Patel and V. Sharma, "Unboxing machine learning models for concrete strength prediction using XAI," *Scientific Reports*, vol. 13, pp. 19023-19038, 2023.

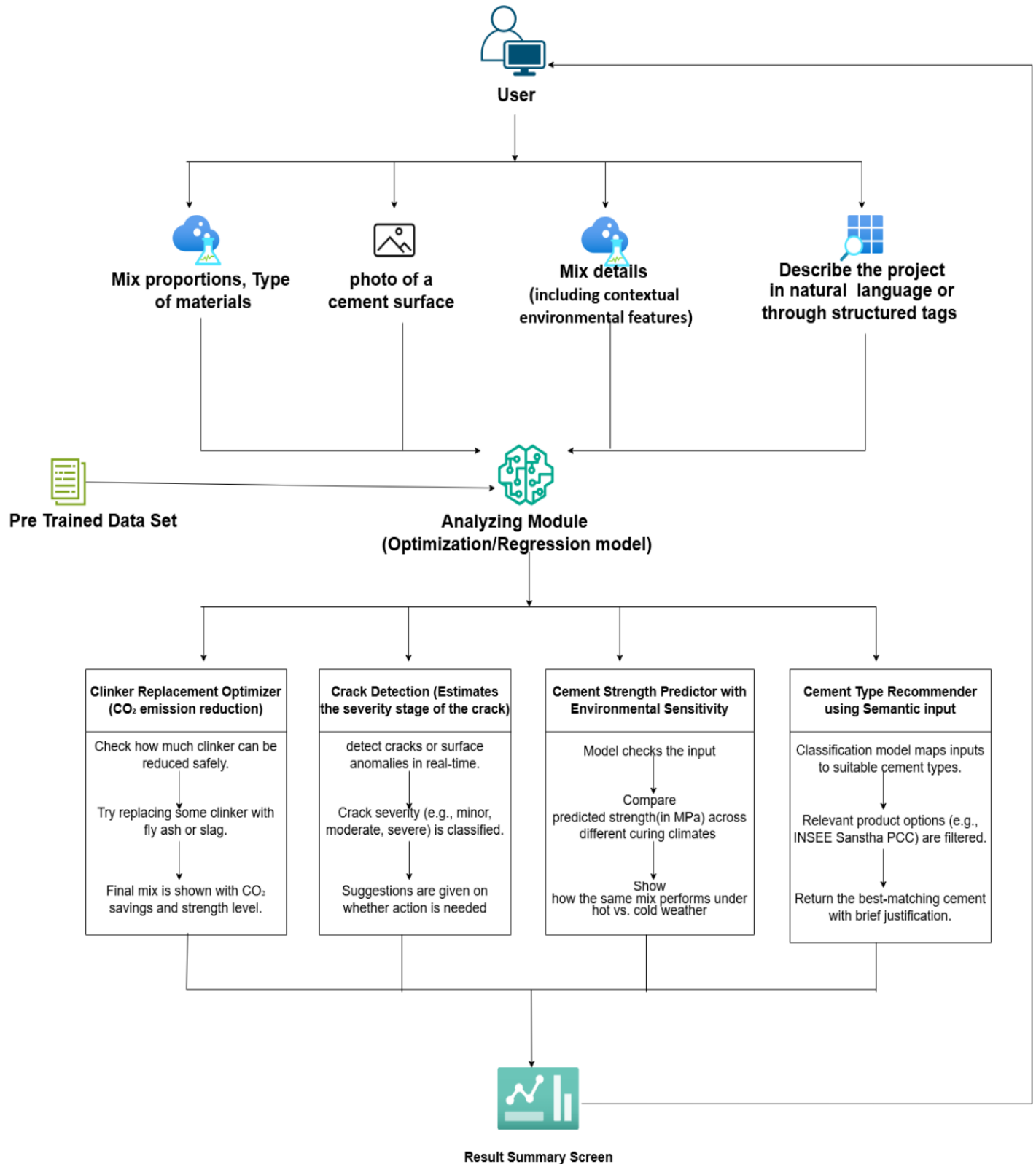
- [10] W. Zhang et al., "Computer vision-based concrete crack identification using MobileNetV2 neural network and adaptive thresholding," *Buildings*, vol. 10, no. 2, pp. 42-58, 2025.
- [11] X. Liu and Y. Park, "A comparative study on crack detection in concrete walls using transfer learning techniques," *Applied Sciences*, vol. 7, no. 4, pp. 169-185, 2023.
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- [13] J. Brown and K. Taylor, "Automated crack detection and crack depth prediction for reinforced concrete structures using deep learning," *Construction and Building Materials*, vol. 370, pp. 130642-130655, 2023.
- [14] N. Garcia et al., "Comparative analysis of deep learning models for crack detection in buildings," *Scientific Reports*, vol. 15, pp. 1245-1262, 2025.
- [15] M. Hassan and O. Ali, "Concrete crack detection using ridgelet neural network optimized by advanced human evolutionary optimization," *Scientific Reports*, vol. 15, pp. 1058-1075, 2025.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The cement industry is a major contributor to CO₂ emissions, primarily due to the high usage of clinker. To address this challenge, a smart and eco-friendly web application is being developed using machine learning. This system aims to suggest optimal material mixes that reduce clinker usage by incorporating alternatives such as fly ash or slag. These recommendations will help minimize emissions while ensuring the quality and strength of the cement remain uncompromised.

The system will provide four main features. Cracks will be detected early by analyzing uploaded photos and recognizing surface textures through computer vision techniques. Cement strength will be predicted using a combination of mix data and environmental factors, including temperature, humidity, and curing time. Construction needs described in plain language will be processed using a semantic input model to recommend the most appropriate cement type. Optimal clinker replacement ratios will be generated using trained machine learning models to support environmentally friendly mix designs.

Through a user-friendly web interface, users will be able to input data, upload images, and receive tailored insights and recommendations. This solution empowers users to make smarter, greener decisions in cement mix design. By reducing reliance on clinker and promoting sustainable material use, the system contributes significantly to lowering CO₂ emissions and advancing the construction industry toward a more sustainable future.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

To build this web application, we identified that we need domain expertise in civil engineering side to understand cement, concrete and construction needs, like how cement powder is making and how much CO₂ emissions happens, how to predict cement strength, what types of cement making, what types of cracks that we can identified and why crack happen. Also we identified that we need Machine Learning (ML) side to create models that can analyze images (for crack detections), predict strength, optimize clinker use, and recommend cement types like that. The Knowledge of web development is essential to make this application user-friendly, allowing users to input data such as photos or mix ratios and get results. Finally, understanding sustainability helps reduce CO₂ emissions by using eco-friendly materials like fly ash or slag. This mix of engineering, AI, and app-building skills ensures that this application solves real-world cement problems effectively.

By understand cement chemistry, like how clinker, water, and additives such as fly ash, slag affect strength and durability. And planning to learn about machine learning basics, such as how to train models to analyze images, predict numbers (like strength), and optimize ratios. And by familiarity with image processing is key for detecting cracks in photos, and construction standards helps recommend the right cement. For a example IS 456 is construction standard for cement types. And using the knowledge of CO₂ emissions in cement production will help to reduce environmental impact. This is combination of cement science, AI, and construction will help to design an application that is accurate and practical.

For Crack Detection, we need 500 -600 labeled concrete images both cracked and non-cracked from cement factory we currently speaking with called INSEE or from Kaggle only if we could not find that amount of images (Concrete Crack and non-Cracked Images). Cement Strength Predictor requires 500–2,000 tabular samples of mix ratios and strength (MPa) from INSEE Lab data and we would be able to find data from online dataset resource called UCI Machine learning. normalized with pandas. Clinker Replacement Optimizer needs 500–1,000 samples of clinker-SCM ratios, strength, and CO₂ emissions from INSEE lab. But In case if couldn't able to get data from the INSEE lab we expect to explore online dataset from Kaggle, Nature Communications Materials refined with Pyomo. Cement Type Recommender uses 200–1,000 samples linking construction needs to cement types (e.g., PPC) from INSEE surveys.

8. Objectives and Novelty

Main Objective Develop an automated system for intelligent cement mix design, aimed at enhancing sustainability, minimizing CO ₂ emissions, and improving the efficiency of cement production and construction processes. By utilizing advanced machine learning algorithms and material characterization techniques, the system seeks to optimize clinker replacement ratios, predict cement strength with high accuracy, detect surface cracks and defects through image analysis, and recommend suitable cement types for various construction applications.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22584472 Koralage K.Y.P	Suggests optimal clinker replacement ratios to reduce CO ₂ emissions.	- Collect a dataset that includes cement composition (clinker, fly ash, slag, etc.) and corresponding strength values. - Preprocess and normalize the dataset to ensure consistency and model readiness. - Label and calculate CO₂ emission values for each cement mix using emission factor data. - Train machine learning models (e.g., Random Forest or XGBoost) to find the best clinker replacement combinations that	A novel system is being proposed to recommend clinker replacement ratios in cement mixes with the aim of reducing CO₂ emissions. While typical approaches prioritize strength alone, this solution is designed to balance environmental impact and performance. Using machine learning models, optimal blends with reduced clinker content are identified, incorporating alternative materials such as fly ash and slag. Emission values are estimated for each mix, allowing greener choices to be highlighted without sacrificing strength. Through this method, more sustainable cement production can be encouraged,

		maintain strength while reducing emissions. Evaluate and validate the model to ensure recommended mixes are both strong and environmentally friendly.	supporting efforts to lower the carbon footprint of the construction industry.
IT22562210 Lakshan V.G.S	Progressive Crack and Severity Classification Module	- Collect and organize crack images over time from various surfaces under different lighting and wear conditions. - Preprocess images to normalize size, remove noise, and enhance crack features using techniques like edge detection or CLAHE. - Train a Machine Learning (ML) model to detect cracks and track their changes over time (progression). - Develop a severity classification system to label cracks (e.g., minor, moderate, severe) based on length, width, and shape.	A progressive crack monitoring module with severity classification is being developed to enhance surface damage detection in cement-based structures. Unlike conventional models that only detect cracks, this system tracks their growth over time and classifies severity based on visual patterns. Temporal image sequences are analyzed using machine learning to assess the rate and extent of development. Severity levels are assigned through predefined thresholds, enabling users to prioritize repairs based on urgency. By combining detection, progression tracking, and risk assessment, this approach offers a more practical and intelligent solution for long-term structural maintenance.

IT22368034 Weerasooriya K H P	Cement Strength Predictor with Environmental Sensitivity	• Add temperature and humidity data to the cement dataset, using real or synthetic values. • Clean and preprocess the dataset for training and testing. • Build a machine learning model using both material mix and environmental inputs. • Evaluate model performance under varying weather scenarios to assess accuracy and generalization. • Visualize results with clear graphs or charts to show prediction accuracy, feature importance, and strength variation across conditions.	A novelty is being created to predict cement strength better than what is done by old ways. Instead of only how materials are mixed being looked at, weather conditions like temperature and humidity are also being considered by this method when concrete is being dried. More accurate predictions are helped to be made by this, especially when work is being done in different places with different weather. By real weather effects being included, better work is expected to be done by the model, more reliable results are expected to be given, and better choices are expected to be helped to be made by people when construction projects are being planned, and cement mixes are being chosen. The environment is considered better by this method than what is done by current models, and more practicality is offered.
IT22640284 Abeywickrama A C A	Recommends the correct cement type for various construction needs	• Collect and categorize data on various construction requirements. • Train classification models to recommend suitable cement types.	Beyond strict form inputs, natural language understanding enables users to freely describe construction contexts. Nuanced scenarios that cannot be fully expressed through fixed criteria are captured through semantic input

		• Build an intuitive user interface for easy recommendations. • Test and validate recommendation accuracy with real-world data. • Integrate a chatbot interface to collect user input and deliver cement recommendations in a conversational and user-friendly manner.	processing. multi-criteria decision-making, based on fuzzy logic, handles ambiguity and imprecision in user input, leading to more accurate and adaptable cement recommendations. Conversational data gathering via a chatbot interface enhances user engagement, streamlines complex input collection, and supports sustainable, context-aware cement decisions by considering multiple structural and environmental factors simultaneously. This approach replaces technical questionnaires with natural, intuitive dialogue, improving accessibility for both experts and non-experts.
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22584472 Koralage K.Y.P	The clinker replacement optimizer is a clever, data-powered tool is built, combining civil engineering and machine learning to recommend cement mixes with less clinker and green additives like fly ash or slag, cutting CO ₂ while ensuring strength. Unlike standard mixes, it uses ML models trained on 500–1,000 Insee (Pvt) Ltd samples of cement blends, strength, and CO ₂ data, processed with Python and optimized using Pyomo. Users input needs, like 40 MPa strength, via an easy interface, and ML predicts the best eco-friendly mix. This reduces CO ₂ by up to 20% with high accuracy, meeting engineering standards.
IT22562210 Lakshan V.G.S	The progressive crack monitoring and severity classification module enhances the assessment of surface cracks in cement-based structures by integrating IT and Data Science. Unlike conventional systems that only detect cracks, this module tracks their progression over time for a comprehensive understanding of structural damage. Temporal image sequences are analyzed using machine learning algorithms to detect subtle changes in crack patterns. Crack features such as length, width, and shape are evaluated, and severity is categorized as minor, moderate, or severe based on predefined criteria. Image data are continuously managed and preprocessed to improve quality, with results shown through an intuitive interface. Machine learning models are trained to accurately monitor progression and classify severity, enabling better prioritization of maintenance activities and improving the safety and durability of cement structures.
IT22368034 Weerasooriya K H P	The cement strength predictor with environmental sensitivity is a smart, data-driven project that brings together the power of Information Technology and Data Science to better predict how strong cement will be. Unlike older models that only look at the mix of materials, this one also considers real-world factors like temperature and humidity during curing, which can have a big impact on the final strength. From an IT angle, the project involves managing and combining different types of data, developing software, and possibly turning it into an easy-to-use application. On the Data Science side, it uses machine learning, carefully chosen features, and model testing to make accurate predictions. In the end, this project shows how technology can help construction professionals make smarter, more informed decisions based on real conditions.

IT22640284 Abeywickrama A C A	As part of a web application, a cement type recommender is implemented using machine learning classification to suggest the ideal cement type based on the user's construction needs. Real-world data from INSEE Cement is used to train the model, factoring in variables like load-bearing, environmental conditions, and moisture exposure. A chatbot interface is integrated to enhance usability, guiding users through a conversational input process that naturally gathers construction details. This makes the recommendation process clear and easy to use, enabling faster, well-informed decisions. By promoting cement choices aligned with industry standards, the system also supports sustainability.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.